comparative -analysis-

How parameters affect on model

	increase	decrease
n_estimators	better accurasy more training time & memory usage	Fast training Less stable prediction
max_depth	Overfitting Captures complex patterns	Underfitting Simple model
min_samples_leaf	Reduce overfitting May underfit	Overfiting More precise splits
random_state	This doesn't change performance, It makes the results repeatable. * Therefore always keep this fixed when comparing	
n_jobs	It will helpto speedup training. There is no effect on accurasy.	

MAE – average absolute difference between actual vs predicted values

* MAE << target range [test]
* test_MAE ≃ Train_MAE , if Test_MAE >> Train_MAE Overfitting

R^2 [coefficient of Determination] - How much variance is explained by the model

R^2	
> 0.9	Excelent
0.7 – 0.9	Good
0.5 – 0.7	Acceptable
0 – 0.5	Weak
< 0	✗

* Train R^2 ↑ but test R^2 ↓ → Overfitting

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

$R^2 = 1$ ✓
$R^2 = 0$ Predict mean
$R^2 < 0$ ✗

MAPE	
< 5%	Excellent
5% – 10%	very good
10% – 20%	Acceptable
> 20%	Poor

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\frac{|y_i - \hat{y}_i|}{|y_i|} \times 100\%$$

target range : (365 - 10052)

n_estimators = 100
max_depth = 8
learning_rate = 0.1
random_seed = 42
verbose = 0

loss function : RMSE

Test_data

Rev. center 1	MAE	R^2	MAPE
Breakfast	679	0.238	∞
Lunch	407	0.217	∞
Dinner	1029	0.358	51.3%

Rev. center 2	MAE	R^2	MAPE
Breakfast	243	0.273	∞
Lunch	464	0.11	∞
Dinner	354	0.195	∞

Rev. center 3	MAE	R^2	MAPE
Breakfast	250	0.066	∞
Lunch	897	0.094	30.2 %
Dinner	3630	0.456	32.9 %

Rev center 4	MAE	R^2	MAPE
Breakfast	—	—	—
Lunch	—	—	—
Dinner	0	0	∞

Rev. center 5	MAE	R^2	MAPE
Breakfast	2542	0.591	16.6%
Lunch	1609	0.1	∞%
Dinner	6447	0.202	79.1%

Rev center 6	MAE	R^2	MAPE
Breakfast	186	0.358	∞%
Lunch	419	0.173	∞%
Dinner	1371	0.212	∞%

Rev center 7	MAE	R^2	MAPE
Breakfast	23	0.332	∞%
Lunch	37	0.163	∞%
Dinner	75	0.09	∞%

Rev center 8	MAE	R^2	MAPE
Breakfast	1981	0.1	∞%
Lunch	7328	0.209	∞%
Dinner	17 458	0.035	∞%

Rev center 9	MAE	R^2	MAPE
Breakfast	218	0.066	∞%
Lunch	147	0.088	∞%
Dinner	100	-0.118	∞%

- conclusion -

MAE(test) should be << 10052 ∴ Rev center 1, 2, 3, 6, 7, 9
not suitable for Rev center 4 (Similar data), 8

suitable for some meal periods in other revenue center

Training data

Rev.center 1	MAE	R^2	MAPE%
Breakfast	217	0.904	∞
Lunch	162	0.902	∞
Dinner	399	0.895	20.2

Rev.center 2	MAE	R^2	MAPE
Breakfast	35	0.811	∞%
Lunch	109	0.878	∞%
Dinner	181	0.888	∞%

Rev.center 3	MAE	R^2	MAPE
Breakfast	104	0.858	∞%
Lunch	349	0.846	30.2%
Dinner	1241	0.92	12.9%

Rev center 4	MAE	R^2	MAPE
Breakfast			
Lunch			
Dinner	0	0.164	∞%

Rev. center 5	MAE	R^2	MAPE
Breakfast	711	0.975	6.2%
Lunch	478	0.944	∞%
Dinner	1334	0.923	56.7%

Rev Center 6	MAE	R^2	MAPE
Breakfast	63	0.885	∞%
Lunch	124	0.879	∞%
Dinner	280	0.974	∞%

Rev center 7	MAE	R^2	MAPE
Breakfast	19	0.829	∞
Lunch	15	0.808	∞
Dinner	21	0.867	∞

Rev center 8	MAE	R^2	MAP
Breakfast	500	0.777	∞%
Lunch	3511	0.862	∞%
Dinner	5093	0.877	∞%

Rev center 9	MAE	R^2	MAPE
Breakfast	63	0.9	∞%
Lunch	61	0.866	∞%
Dinner	35	0.811	∞%

* This model is very suitable for revcenter 1, 2 [both MAE equals and small), rev_center 6,7 (small and relatively equal) and 9 [small and relatively equal),

target range 365.5 – 10052

n_estimators = 200
max_depth = 0.05
learning_rate = 10
random_seed = 60
verbose = 0

loss function = RMSE

Test_data

Rev. center 1	MAE	R^2	MAPE
Breakfast	683	0.224	∞%
Lunch	399	0.307	∞%
Dinner	1002	0.430	∞%

Rev. center 2	MAE	R^2	MAPE
Breakfast	251	0.254	∞%
Lunch	457	0.083	∞%
Dinner	372	0.149	∞%

Rev. center 3	MAE	R^2	MAPE
Breakfast	256	0.009	∞%
Lunch	926	−0.025	58.9%
Dinner	3731	0.438	35.5%

Rev center 4	MAE	R^2	MAPE
Breakfast			
Lunch			
Dinner	0	0	∞%

Rev. center 5	MAE	R^2	MAPE
Breakfast	2604	0.577	17.8%
Lunch	1683	0.055	∞%
Dinner	6555	0.18	69.5%

Rev center 6	MAE	R^2	MAPE
Breakfast	192	0.295	∞%
Lunch	425	0.145	∞%
Dinner	1366	0.122	∞%

Rev center 7	MAE	R^2	MAPE
Breakfast	23	0.299	∞%
Lunch	37	0.168	∞%
Dinner	77	0.059	∞%

Rev center 8	MAE	R^2	MAPE
Breakfast	1977	0.098	∞%
Lunch	7084	0.223	∞%
Dinner	17233	0.014	∞%

Rev center 9	MAE	R^2	MAPE
Breakfast	216	0.067	∞%
Lunch	146	0.061	∞%
Dinner	92	−0.018	∞%

suitable for :- Rev center 1, 2, 3, 6, 7, 9 & some meal periods for other centers.

training data

Rev. center 1	MAE	R^2	MAPE
Breakfast	190	0.926	∞%
Lunch	126	0.942	∞%
Dinner	325	0.931	16.5%

Rev. center 2	MAE	R^2	MAPE
Breakfast	75	0.924	∞%
Lunch	84	0.931	∞%
Dinner	148	0.926	∞%

Rev. center 3	MAE	R^2	MAPE
Breakfast	79	0.92	∞%
Lunch	270	0.908	23.9%
Dinner	1027	0.945	10.8%

Rev center 4	MAE	R^2	MAPE
Breakfast			
Lunch			
Dinner	0	0.164	∞%

Rev. center 5	MAE	R^2	MAPE
Breakfast	627	0.981	44%
Lunch	372	0.966	∞%
Dinner	1040	0.954	33.3%

Rev center 6	MAE	R^2	MAPE
Breakfast	46	0.94	∞%
Lunch	99	0.921	∞%
Dinner	254	0.977	∞%

Rev center 7	MAE	R^2	MAPE
Breakfast	14	0.898	∞%
Lunch	14	0.852	∞%
Dinner	17	0.919	∞%

Rev center 8	MAE	R^2	MAPE
Breakfast	455	0.8	∞%
Lunch	2911	0.903	∞%
Dinner	4225	0.92	∞%

Rev center 9	MAE	R^2	MAPE
Breakfast	51	0.935	∞%
Lunch	46	0.926	∞%
Dinner	27	0.876	∞%

This method is very suitable for 1, 2, 7, 9 target range 365 – 10052

n_estimators = 300
max_depth = 10
learning_rate = 0.01
random_seed = 85
verbose = 0

loss function = RMSE

Test_data

Rev. center 1	MAE	R^2	MAPE
Breakfast	642	0.262	∞%
Lunch	412	0.283	∞%
Dinner	968	0.463	50.8%

Rev. center 2	MAE	R^2	MAPE
Breakfast	258	0.196	∞%
Lunch	447	0.086	∞%
Dinner	375	0.146	∞%

Rev. center 3	MAE	R^2	MAPE
Breakfast	253	0.062	∞%
Lunch	855	0.192	57%
Dinner	3632	0.428	34.9%

Rev center 4	MAE	R^2	MAPE
Breakfast			
Lunch			
Dinner	0	0	∞%

Rev. center 5	MAE	R^2	MAPE
Breakfast	2563	0.628	17.9%
Lunch	1661	0.074	∞%
Dinner	6560	0.137	67%

Rev center 6	MAE	R^2	MAPE
Breakfast	195	0.258	∞%
Lunch	436	0.138	∞%
Dinner	1316	0.108	∞%

Rev center 7	MAE	R^2	MAPE
Breakfast	25	0.292	∞%
Lunch	36	0.17	∞%
Dinner	76	0.071	∞%

Rev center 8	MAE	R^2	MAPE
Breakfast	1733	0.119	∞%
Lunch	7010	0.199	∞%
Dinner	17303	0.011	∞%

Rev center 9	MAE	R^2	MAPE
Breakfast	211	0.118	∞%
Lunch	148	0.087	∞%
Dinner	85	-0.008	∞%

suitable for :- Rev center 1, 2, 3, 6, 7, 9 some meal periods for
other centers.

training data

Rev. center 1	MAE	R^2	MAPE
Breakfast	309	0.744	∞ %
Lunch	253	0.722	∞ %
Dinner	576	0.772	28 %

Rev. center 2	MAE	R^2	MAPE
Breakfast	146	0.68	∞ %
Lunch	170	0.671	∞ %
Dinner	255	0.663	∞ %

Rev. center 3	MAE	R^2	MAPE
Breakfast	164	0.638	∞ %
Lunch	477	0.678	40.6 %
Dinner	1721	0.83	17.6 %

Rev center 4	MAE	R^2	MAPE
Breakfast			
Lunch			
Dinner	0	0.156	∞ %

Rev. center 5	MAE	R^2	MAPE
Breakfast	1228	0.916	7.7 %
Lunch	790	0.803	∞ %
Dinner	2099	0.774	60.7 %

Rev center 6	MAE	R^2	MAPE
Breakfast	93	0.697	∞ %
Lunch	175	0.704	∞ %
Dinner	444	0.726	∞ %

Rev center 7	MAE	R^2	MAPE
Breakfast	25	0.292	∞ %
Lunch	22	0.563	∞ %
Dinner	32	0.678	∞ %

Rev center 8	MAE	R^2	MAPE
Breakfast	716	0.551	∞ %
Lunch	4993	0.688	∞ %
Dinner	7816	0.71	∞ %

Rev center 9	MAE	R^2	MAPE
Breakfast	97	0.724	∞ %
Lunch	97	0.654	∞ %
Dinner	48	0.618	∞ %

This method is very suitable for 1, 2, 6, 7, 9

Final conclusion

Random Forest method is very suitable for predict future revenue on revenue center 1, 2, 6, 7, 9